

1. Research Problem & Scientific Thesis

Operator Learning on Irregular Topologies

SeismoFNO investigates whether neural operators can learn seismic structural response while generalizing across both structural topology and modal-period distribution shift. Predicting non-linear dynamic responses of multi-story buildings subjected to transient ground motions requires solving non-linear differential systems currently evaluated through high-fidelity numerical integration in OpenSeesPy. While surrogate neural operators offer large evaluation acceleration, canonical architectures exhibit critical physical and geometric failure modes when deployed across heterogeneous civil structures.

Through systematic research progression across **2,160 physical simulations**, we establish: **(1) EXP4:** Conventional fixed-grid 2D Fourier Neural Operators (FNO2D) fail catastrophically when applied to variable-story structures because spatial zero-padding introduces artificial discontinuities (**99.60% median Rel. L2 error** on 3-story frames); **(2) EXP5:** A topology-native Graph Neural Operator (GNO) resolves boundary artifacts by operating exclusively on physical floor nodes and column edges (reducing 3-story error to **22.09%**), yet suffers severe phase opposition under structural period extrapolation; **(3) EXP6:** Conditioning spatiotemporal operator blocks on structural eigenvalue invariants via Feature-wise Linear Modulation (FiLM) substantially improves peak-response generalization under structural modal-period shift (reducing unseen archetype peak error from **35.21% to 13.06%**). **Limitation:** Trajectory-level waveform fidelity remains limited under strong modal extrapolation because global 1D Fourier layers accumulate temporal phase drift over long horizons.

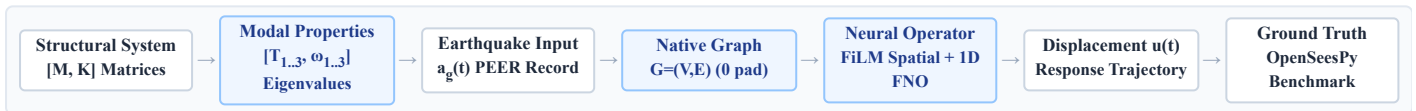
2. Governing Dynamics & Computational Pipeline

Physics Ground Truth: OpenSeesPy NLTHA

The surrogate approximates the structural response generated by non-linear time-history analysis (NLTHA) under earthquake ground excitation. Multi-degree-of-freedom (MDOF) shear buildings subjected to horizontal base acceleration $a_g(t)$ are governed by:

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{K} \mathbf{u}(t) = -\mathbf{M} \mathbf{r} a_g(t) \quad (1) \text{ Coupled Matrix Dynamic Equation}$$

where $\mathbf{M}, \mathbf{C}, \mathbf{K} \in \mathbb{R}^{N \times N}$ denote mass, Rayleigh damping, and lateral stiffness matrices, $\mathbf{r} = [1, \dots, 1]^T$ is the structural influence vector, and $\mathbf{u}(t) \in \mathbb{R}^N$ is the relative floor displacement trajectory across duration $t \in [0, 20.48]$ s.



3. Four-Stage Research Progression (EXP4 → EXP5 → EXP6)

Systematic Hypothesis Testing & Failure Analysis

EXP4: BASELINE FAILURE Fixed-Grid 2D FNO Standard 2D FNO on fixed 5×2048 grid. 3-story buildings zero-padded at stories 4–5. Fourier basis enforces non-physical spatial periodicity across zero boundary. 3-Story Rel L2: 99.60% (Fail)	EXP5: TOPOLOGY NATIVE Spatiotemporal GNO Replaced Cartesian grid with physical graph $G=(V,E)$. Zero artificial padding. Eliminates spatial ringing but reveals temporal phase divergence under modal shift. 3-Story Rel L2: 22.09% (Resolved)	EXP6: PHYSICS FILM Modal-Conditioned GNO Injected pre-earthquake structural invariants $[T_1, \omega_1]$ into spatial message-passing and spectral layers via FiLM scale/shift generators. OOD-B Peak Error: 13.06%	ACTIVE LIMITATION Phase Drift Over Horizon Global 1D Fourier layers lack dynamic frequency warping. While peak envelopes match (13.06%), temporal phase divergence accumulates over 20.48 s. OOD-B Rel L2: 124.07% (r≈0.07)
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4. Master Empirical Comparison Across Architectural Phases

Reconstructed from 2,160 Frozen Simulation Evaluations

Experiment	Representation	Scientific Contribution	Key Result
EXP4	Fixed-grid FNO2D (5×2048)	Tests conventional fixed-grid neural operator on variable-floor geometry	3-story median Relative L2 error: 99.60%
EXP5	Topology-native GNO	Removes artificial zero-padding and supports variable topology	3-story median Relative L2 error: 22.09%
EXP6-B	T1-conditioned GNO	Physics-informed FiLM conditioning on fundamental vibration period T_1	OOD-B peak displacement error: 13.47%
EXP6-C	Multi-modal GNO	Conditions on $[T_1-3, \omega_1-3]$ via dual-branch FiLM modulation	OOD-B peak displacement error: 13.06%
EXP6-D	Shuffled T_1 control	Tests whether conditioning benefit depends on physical correspondence	OOD-B peak error: 24.33%

Note on EXP6-D Falsification Control: Provides evidence that performance gains depend on physically meaningful modal correspondence rather than merely additional conditioning capacity (OOD-B peak error degrades from 13.06% to 24.33%, and OOD-C degrades from 17.01% to 36.16%).

STRUCTURAL OOD — 5S_T120 (T1 = 1.20 s)

Training Support: T1 ∈ [0.35, 0.85] s

This structure is outside the training modal-period support. Civil engineers evaluate safety envelopes around peak floor displacement.

EXP5 GNO 35.21% Median Peak Error	EXP6-B 13.47% Median Peak Error	EXP6-C 13.06% Median Peak Error	RELATIVE REDUCTION 62.9% Peak Error Drop
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The Multi-Modal GNO reduces the baseline OOD-B peak displacement error from 35.21% to 13.06%, corresponding to a 62.9% relative reduction under the evaluated structural OOD condition.

6. Computational Result

Measured single-simulation wall-clock comparison on Apple Silicon MPS:

OpenSeesPy NLTHA 54.68 ms Nonlinear Analysis	EXP6 T1-GNO 21.45 ms MPS GPU Forward	Wall-Clock Ratio 2.55× vs OpenSeesPy
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Wall-clock ratio: 2.55× relative to the OpenSeesPy benchmark. Attribution: Measured single-simulation wall-clock comparison on Apple Silicon MPS.

7. Where the Model Still Fails

- 1. Long-horizon waveform phase drift: Relative L2 > 100% under strong modal extrapolation. Pearson correlation approximately 0.05–0.09 in the reported OOD waveform cases. The model can reproduce peak-response envelopes substantially better than full trajectory phase under extrapolation, while global 1D Fourier layers accumulate phase discrepancy over the 20.48 s horizon.
- 2. Static modal descriptors under severe yielding: The modal descriptors are computed from pre-earthquake structural [M] and [K]. They do not explicitly represent dynamic period elongation during severe nonlinear yielding.
- 3. Structural idealization: The current study uses planar lumped-mass shear-frame representations and does not yet cover full 3D asymmetric/torsional/bidirectional structural response.

8. Relevance to CSE & Scientific ML

- Operator learning on dynamic graphs: Demonstrates how to couple discrete spatial message-passing with continuous 1D temporal spectral convolutions for multi-scale dynamical physical systems.
- Disentangled error modes in dynamical systems: Empirically reveals that modal conditioning can repair amplitude/envelope generalization while leaving global temporal phase drift uncorrected.
- Scientific ML software engineering: Enforces cryptographic reproducibility, zero-leakage hash verification, and automated forensic auditing across 2,160 physical simulations.

9. Logical Next Research Direction

Causal Structured State-Space Models (SSMs / S4 / Mamba):

To eliminate the long-horizon phase divergence inherent to static global Fourier spectral layers, next-phase research will investigate causal State-Space Models (SSMs) or Continuous-Time Neural ODEs coupled with topology graphs. A recurrent state-space formulation processes time causally, enabling dynamic stiffness degradation and period elongation to be tracked recursively step-by-step rather than forced onto a static Fourier basis.

10. Reproducibility & Forensic Audit Verification

Zero Data Leakage & Frozen Core

Research core: EXP4 / EXP5 / EXP6 frozen	Data leakage: 0 pairwise partition intersections	Scaler fitting: Training partition only	Modal conditioning: Derived from structural [M], [K] before earthquake excitation
Repository tests: 294 passed, 2 skipped, 0 failed	Historical experiment artifacts: SHA256 integrity verified	EXP7: Not started	Demonstration layer: Audited and frozen

11. RESEARCH RESOURCES

Genuinely Clickable Repository Artifacts

Technical Report (18-Section Formal Report) • Professor Demo Documentation • Independent Forensic Audit • Architecture Specification • Repository README • Interactive Demo — local / repository deployment